Reed Goldsworthy

Reedgoldsworthy2@gmail.com reedgoldsworthy.github.io/ 331-422-8833

EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science, Computer Science + Mathematics

2021 – current, GPA: 3.58 Champaign, IL

Awards: Math Excellence Award; Phi Theta Kappa Honor Society; Dean's List

SKILLS

C++; Python; Git; C; Java; R; HTML; CSS; Microsoft Excel; PyCharm; Microsoft Visual Studio

PROJECTS

Sorting Algorithm Visualizer | Creator | Website: https://bit.ly/3s9x9jF

- Developed a sorting algorithm visualizer web application using HTML, CSS, and JavaScript.
- Implemented various sorting algorithms, including Bubble Sort, Merge sort, and Quick Sort to visually demonstrate the sorting process in real-time.
- Designed an intuitive user interface that allows users to customize the size, speed and type of sort used in the animation.

Real-Time Driving Lane Detection | Creator | GitHub: https://bit.ly/3ORemRZ

- Developed a python program using OpenCV to detect lanes on a road from a video.
- Incorporated Image Masking, Image Thresholding, and Hough Line Transformations to accurately identify road markings at each frame.

PhotoMosaic Maker | Creator | GitHub: https://bit.ly/45qaNc4

- Created a python program using OpenCV to create a photo mosaic of an image from a folder of sub-images.
- Utilized concepts like caching and pre-computation in order to optimize the program and save on computational costs.

Monopoly Simulator | Creator | GitHub: http://bit.ly/3qOVNFN

- Created a python program to simulate games of Monopoly. This project tracks data across multiple games and displays the simulator's findings.
- Simulates over 12,000,000 turns/100,000 games.
- Utilized Matplotlib in order to clearly visualize the simulator's findings.

RELEVANT COURSES

- Data Structures
- Computer Architecture
- Intro to Algorithms & Models of Computation
- Statistics & Probability
- System Programming
- Numerical Analysis
- Discrete Structures
- Linear Algebra w/ Computation Applications